

Safety Data Sheet

OSHA HazCom Standard 29 CFR 1910.1200(g) revised in 2024 and GHS Rev 07

Date of issue: 06/23/2025

Revised on 06/23/2025

* 1 Identification

· Product Identifier

· **Trade Name:** Bare Stainless Steel Welding Electrodes and Rods

· **Other means of identification**

· **Product Number:**

Specification: A5.9

Classification: ER16-8-2, ER2209, ER2594, ER307, ER308H, ER308L, ER308LSi, ER309H, ER309LSi, ER310, ER312, ER316H, ER316L, ER316LSi, ER317L, ER320LR, ER347, ER385, ER409Nb, ER410, ER420, ER430, ER630

Bare stainless steel welding electrodes and rods

Specification: None

Classification: ER307Si, ER309LMo (ISO 14343-A; 23 12 2 L), ER316MnNF, ER330, ER410NiMo

Bare stainless steel welding electrodes and rods

· **Relevant identified uses of the substance or mixture and uses advised against:**

For professional use only. Use according to manufacturer's specification.

· **Product Description:** Bare stainless steel welding electrodes and rods welding rods and electrodes.

· **Application of the substance / the mixture:** Industry specific application.

· **Details of the Supplier of the Safety Data Sheet:**

· **Manufacturer/Supplier:**

Pinnacle Alloys I, LLC

9384 Wallisville Road

Houston, TX 77013

Telephone: 800-856-9353

· **Emergency telephone number:** 713-688-9353

* 2 Hazard(s) Identification

· **Classification of the substance or mixture:**



Health hazard

Sensitization - respiratory 1

H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.

Carcinogenicity 2

H351 Suspected of causing cancer. Route of exposure: Inhalation.

Specific target organ toxicity (repeated exposure) 1

H372 Causes damage to the respiratory system through prolonged or repeated exposure. Route of exposure: Inhalation.



Skin irritation 2

H315 Causes skin irritation.

Eye irritation 2A

H319 Causes serious eye irritation.

Sensitization - skin 1

H317 May cause an allergic skin reaction.

Specific target organ toxicity (single exposure) 3

H335 May cause respiratory irritation.

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Trade Name: Bare Stainless Steel Welding Electrodes and Rods

- **Label elements:**
- **Hazard pictograms:**



- **Signal word:** Danger
- **Hazard-determining components of labeling:**

Nickel
Iron
Cobalt
copper

- **Hazard statements:**

H315 Causes skin irritation.
H319 Causes serious eye irritation.
H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317 May cause an allergic skin reaction.
H351 Suspected of causing cancer. Route of exposure: Inhalation.
H335 May cause respiratory irritation.
H372 Causes damage to the respiratory system through prolonged or repeated exposure. Route of exposure: Inhalation.

- **Precautionary statements:**

P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P260 Do not breathe dust/fume/gas/mist/vapors/spray.
P264 Wash thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P272 Contaminated work clothing must not be allowed out of the workplace.
P280 Wear protective gloves/protective clothing/eye protection/face protection/hearing protection.
P284 [In case of inadequate ventilation] wear respiratory protection.
P302+P352 If on skin: Wash with plenty of water.
P304+P340 If inhaled: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338 If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308+P313 IF exposed or concerned: Get medical advice/attention.
P312 Call a poison center/doctor if you feel unwell.
P321 Specific treatment (see supplementary first aid instructions on this Safety Data Sheet).
P362+P364 Take off contaminated clothing and wash it before reuse.
P333+P313 If skin irritation or rash occurs: Get medical advice/attention.
P337+P313 If eye irritation persists: Get medical advice/attention.
P342+P311 If experiencing respiratory symptoms: Call a poison center/doctor.
P403+P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.
P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

- **Unknown acute toxicity:**

This value refers to knowledge of known, established toxicological or ecotoxicological values.
40.8 % of the mixture consists of component(s) of unknown toxicity.

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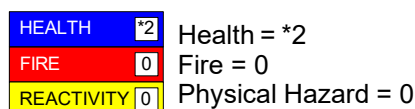
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Trade Name: Bare Stainless Steel Welding Electrodes and Rods

- **Information pertaining to particular dangers for man and environment:**
- **Classification system:** NFPA/HMIS Definitions: 0-Least, 1-Slight, 2-Moderate, 3-High, 4-Extreme
- **NFPA ratings (scale 0 - 4)**



- **HMIS-ratings (scale 0 - 4)**



- **Hazard(s) not otherwise classified (HNOC):** None known
- **Classification according to (d)(1)(ii) of § 1910.1200**
The SDS issuer does not object to the classifications provided by importers or manufacturers of precursor products.
- **Hazards not otherwise classified**
There are no adverse physical or health effects known that are not covered by the hazard classes of the Hazard Communications Standard.

* 3 Composition/Information on Ingredients

- **Chemical characterization: Substance**
- **Description:**

Mixture of substances listed with non-hazardous additions and/or components that are less than listing threshold. All non-hazardous, components listed under threshold, and listed component total 100%.

- **Dangerous Components:**

CAS: 7439-89-6 RTECS: NO 4565500	Iron ⚠ Flammable solids 2, H228; ⚠ Skin irritation 2, H315; Specific target organ toxicity (single exposure) 3, H335; Eye irritation 2B, H320; Combustible Dust	Balance%
CAS: 7440-47-3 RTECS: GB 4200000	Chromium	<40%
CAS: 7440-02-0	Nickel ⚠ Carcinogenicity 2, H351; Specific target organ toxicity (repeated exposure) 1, H372; ⚠ Sensitization - skin 1, H317; Aquatic Acute 3, H402	<46%
CAS: 7439-96-5 RTECS: OO 9275000	Manganese ⚠ Pyrophoric solids 1, H250; Substances and mixtures which, in contact with water, emit flammable gases 1, H260	<15%
CAS: 7439-98-7 RTECS: QA 4680000	Molybdenum	<7%
CAS: 7440-50-8 RTECS: GL 5325000	copper ⚠ Aquatic Acute 1, H400 (M=10)	<4%
CAS: 7440-21-3	Silicon ⚠ Flammable solids 2, H228; ⚠ Acute toxicity - oral 4, H302; Eye irritation 2B, H320; Combustible Dust	<3.3%
CAS: 7440-33-7 RTECS: YO 7175000	Tungsten ⚠ Flammable solids 1, H228; ⚠ Acute toxicity - oral 4, H302; Acute toxicity - dermal 4, H312; Skin irritation 2, H315; Eye irritation 2A, H319	<1.5%

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CAS: 7440-03-1 RTECS: QT9900000	Niobium ⚠ Flammable solids 1, H228; Combustible Dust	<1.35%
CAS: 7440-48-4 RTECS: GF 8750000	Cobalt ⚠ Sensitization - respiratory 1, H334; Germ cell mutagenicity 2, H341; Carcinogenicity 1B, H350; Reproductive toxicity 1B, H360; ⚠ Sensitization - skin 1, H317; Combustible Dust	<1%
CAS: 7440-44-0 RTECS: FF 5250100	Carbon	<1%

· **Additional information:**

The exact percentages of the ingredients of this mixture are considered to be proprietary and are withheld in accordance with the provisions of paragraph (i) of §1910.1200 of 29 CFR 1910.1200 Trade Secrets.

Note: Certain chemical constituents listed in Section 3 may vary depending upon the Classification of the Bare Stainless Steel Welding Electrodes and Rods products.

* 4 First-Aid Measures

· **Description of first aid measures**

· **General information:**

Symptoms of poisoning may occur after exposure to dust, fumes or particulates; seek medical attention if feeling unwell.

· **After inhalation:**

Supply fresh air. If required, provide artificial respiration. Consult doctor if symptoms persist. In case of unconsciousness, place patient stably in side position for transportation.

· **After skin contact:**

Immediately wash with water and soap and rinse thoroughly.

If skin irritation occurs, consult a doctor.

· **After eye contact:**

Do NOT rub eyes. Immediately rinse opened eye(s) for at least 15 minutes under running water, lifting upper and lower lids occasionally. If symptoms persist, consult a physician.

Rinse opened eye for at least 15 minutes under running water. If symptoms persist, consult a doctor.

If eye irritation occurs, consult a doctor.

· **After swallowing:**

Rinse out mouth and then drink plenty of water.

Do not induce vomiting without medical advice.

If swallowed and symptoms occur, consult a doctor.

· **Most important symptoms and effects, both acute and delayed:** No further relevant information available.

· **Indication of any immediate medical attention and special treatment needed:**

No further relevant information available.

* 5 Fire-Fighting Measures

· **Extinguishing media**

· **Suitable extinguishing agents:**

CO₂, extinguishing powder or water spray.

Use fire fighting measures that suit the environment.

· **For safety reasons unsuitable extinguishing agents:** No further relevant information.

· **Special hazards arising from the substance or mixture:**

Amorphous or crystalline silicon both react exothermically when heated with alkali-metal carbonates attaining incandescence and evolving carbon monoxide.

Material in powder form, capable of creating a dust explosion. Mixture of silicon, aluminum, and lead oxide explodes when heated.

Moderate fire hazard when it is in the form of a dust (powder) and burns rapidly when heated in flame. Chromium is attacked vigorously by fused potassium chlorate producing vivid incandescence. Pyrophoric chromium unites with nitric oxide with incandescence. Incandescent reaction with nitrogen oxide or sulfur

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dioxide.

Special Remarks on Explosion Hazards:

Powdered Chromium metal +fused ammonium nitrate may react violently or explosively. Powdered Chromium will explode spontaneously in air.

Material in powder form is capable of creating a dust explosion. Mixture of silicon, aluminum, and lead oxide explodes when heated.

If incinerated, product will release the following toxic fumes: Oxides of carbon, chromium, copper, iron, manganese, molybdenum, nickel, niobium, silicon, tungsten and cobalt.

· **Advice for firefighters**

· **Special protective equipment for firefighters:**

As in any fire, wear self-contained breathing apparatus pressure-demand (NIOSH approved or equivalent) and full protective gear to prevent contact with skin and eyes.

· **Additional information:**

These items are not reactive, flammable, or explosive and essentially not hazardous at ambient temperatures. Welding arcs and sparks can ignite combustibles and flammable products. If involved in a fire, these products may generate irritating aluminum fumes and a variety of metal oxides. Emergency responders must wear personal protection equipment suitable for the situation. Use the extinguishing media recommended for the burning materials and fire situation. See ANSI Z49.1 "Safety in Welding and Cutting" and "Safe Practices" Code: SP, published by the American Welding Society.

* 6 Accidental Release Measures

· **Personal precautions, protective equipment and emergency procedures:**

Ensure adequate ventilation.

Wear protective equipment.

Keep unprotected persons away.

Avoid contact with skin, eyes and clothing.

· **Environmental precautions:** No special measures required.

· **Methods and material for containment and cleaning up:**

Dispose of contaminated material as waste according to section 13.

Ensure adequate ventilation.

Dispose of the collected material according to regulations.

Flammable solid. Stop leak if without risk. Do not touch spilled material. Use water spray curtain to divert vapor drift. Prevent entry into sewers, basements or confined areas; dike if needed. Eliminate all ignition sources.

· **PAC-1:**

7439-89-6	Iron	1.8 mg/m ³
7440-47-3	Chromium	1.5 mg/m ³
7440-02-0	Nickel	0.24 mg/m ³
7439-96-5	Manganese	3 mg/m ³
7439-98-7	Molybdenum	30 mg/m ³
7440-50-8	copper	3 mg/m ³
7440-21-3	Silicon	45 mg/m ³
7440-33-7	Tungsten	10 mg/m ³
7440-03-1	Niobium	30 mg/m ³
7440-44-0	Carbon	6.0 mg/m ³
7440-48-4	Cobalt	0.18 mg/m ³

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· PAC-2:		
7439-89-6	Iron	20 mg/m ³
7440-47-3	Chromium	83 mg/m ³
7440-02-0	Nickel	50 mg/m ³
7439-96-5	Manganese	5 mg/m ³
7439-98-7	Molybdenum	330 mg/m ³
7440-50-8	copper	33 mg/m ³
7440-21-3	Silicon	100 mg/m ³
7440-33-7	Tungsten	330 mg/m ³
7440-03-1	Niobium	330 mg/m ³
7440-44-0	Carbon	330 mg/m ³
7440-48-4	Cobalt	2 mg/m ³
· PAC-3:		
7439-89-6	Iron	120 mg/m ³
7440-47-3	Chromium	500 mg/m ³
7440-02-0	Nickel	99 mg/m ³
7439-96-5	Manganese	500 mg/m ³
7439-98-7	Molybdenum	2,000 mg/m ³
7440-50-8	copper	200 mg/m ³
7440-21-3	Silicon	630 mg/m ³
7440-33-7	Tungsten	2,000 mg/m ³
7440-03-1	Niobium	2,000 mg/m ³
7440-44-0	Carbon	2,000 mg/m ³
7440-48-4	Cobalt	20 mg/m ³

· Reference to other sections:

See Section 7 for information on safe handling.

See Section 8 for information on personal protection equipment.

See Section 13 for disposal information.

* 7 Handling and Storage

· Precautions for safe handling:

Open and handle receptacle with care.

Ensure good ventilation/exhaustion at the workplace.

· Information about protection against explosions and fires: No special measures required.**· Conditions for safe storage, including any incompatibilities**

Store away from strong acids, strong bases, strong oxidizing agents and strong reducing agents.

· Storage**· Requirements to be met by storerooms and receptacles:** Store in the original container.**· Information about storage in one common storage facility:** Not required.**· Further information about storage conditions:** Keep receptacle tightly sealed.**· Specific end use(s):** No further relevant information available.

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* 8 Exposure Controls/Personal Protection

· **Control parameters:**

All ventilation should be designed in accordance with OSHA standard (29 CFR 1910.94). Use local exhaust at filling zones and where leakage and dust formation is probable. Use mechanical (general) ventilation for storage areas. Use appropriate ventilation as required to keep Exposure Limits in Air below TLV & PEL limits.

· **Components with occupational exposure limits:**

The following constituents are the only constituents of the product which have a PEL, TLV or other recommended exposure limit.

At this time, the other constituents have no known exposure limits.

7440-47-3 Chromium

PEL	Long-term value: 1 mg/m ³
REL	Long-term value: 0.5* mg/m ³ *metal+inorg.compds.as Cr; See Pocket Guide App. C
TLV	Long-term value: 0.5* mg/m ³ *metal

7440-02-0 Nickel

PEL	Long-term value: 1 mg/m ³
REL	Long-term value: 0.015 mg/m ³ as Ni; See Pocket Guide App. A
TLV	Long-term value: 1.5* mg/m ³ elemental, *inhalable fraction, A5, BEI

7439-96-5 Manganese

PEL	Ceiling limit value: 5 mg/m ³ as Mn
REL	Short-term value: 3 mg/m ³ Long-term value: 1 mg/m ³ fume, as Mn
TLV	Long-term value: 0.02* 0.1** mg/m ³ as Mn; A4, *respirable **inhalable fraction

7439-98-7 Molybdenum

PEL	Long-term value: 15* mg/m ³ *Total dust, as Mo
TLV	Long-term value: 10* 3** mg/m ³ as Mo; *inhalable fraction **respirable fraction

7440-50-8 copper

PEL	Long-term value: 1* 0.1** mg/m ³ as Cu *dusts and mists **fume
REL	Long-term value: 1* 0.1** mg/m ³ as Cu *dusts and mists **fume
TLV	Long-term value: 0.2* 1** mg/m ³ as Cu *fume **dust/mist

7440-21-3 Silicon

PEL	Long-term value: 15* 5** mg/m ³ *total dust **respirable fraction
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REL	Long-term value: 10* 5** mg/m ³ *total dust **respirable fraction
TLV	TLV withdrawn
7440-33-7 Tungsten	
REL	and insoluble compounds, as We
REL	Short-term value: 10 mg/m ³ Long-term value: 5 mg/m ³ as W
TLV	Long-term value: 3* mg/m ³ as W; * respirable fraction
7440-03-1 Niobium	
TWA	Long-term value: 6
7440-48-4 Cobalt	
REL	Long-term value: 0.1* mg/m ³ as Co; *for metal dust and fume
REL	Long-term value: 0.05 mg/m ³ as Co; metal dust & fume
TLV	Long-term value: 0.02* 0.005** mg/m ³ RSEN,DSEN,A3,BE1,**thoracic particulates,A2
Ingredients with biological limit values:	
7440-47-3 Chromium	
BEI	0.7 µg/L urine end of shift, end of workweek Total chromium (population based)
7440-02-0 Nickel	
BEI	5 µg/L urine post-shift at end of workweek Nickel (background) 30 µg/L urine post-shift at end of workweek Nickel (background)
7440-48-4 Cobalt	
BEI	15 µg/L urine end of shift at end of workweek Cobalt (nonspecific)

· **Additional information:** The lists that were valid during the creation of this SDS were used as basis.

· **Exposure controls:**

· **Appropriate engineering controls** No further data; see section 7.

· **Personal protective equipment**

· **General protective and hygienic measures:**

Keep away from foodstuffs, beverages and feed.

Immediately remove all soiled and contaminated clothing and wash before reuse.

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Trade Name: Bare Stainless Steel Welding Electrodes and Rods

Wash hands before breaks and at the end of work.

Avoid contact with the eyes and skin.

· **Breathing equipment:**



Suitable respiratory protective device recommended.

Use NIOSH approved or equivalent fume respirator or air supplied respirator when welding, brazing, cutting, grinding, or soldering in a confined space or general work area where local exhaust and/or ventilation does not keep exposure below the limits outlined in Section 8. Monitor the air quality inside the welder's helmet, and/or worker's breathing zone to determine if a respirator is required and the type needed.

· **Protection of hands:**

Due to missing tests no recommendation to the glove material can be given for the product/ the preparation/ the chemical mixture.

Select glove material based on penetration times, rates of diffusion and degradation.



Protective gloves

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation.

· **Material of gloves:**

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer. As the product is a preparation of several substances, the resistance of the glove material cannot be calculated in advance and has therefore to be checked prior to the application.

· **Penetration time of glove material:**

The exact break-through time has to be determined and observed by the manufacturer of the protective gloves.

· **Eye protection:**



Helmet or face shield

Wear a helmet or face shield with a filter lens around shade number 14. Adjust if needed by selecting the next lighter or darker shade number. See ANSI/ASC Z49.1 Section 4.2 or publication F2.2. Shield other workers by providing screens and flash goggles.

· **Body protection:**

Wear approved head, hand, and body protection, which help to prevent injury from radiation, sparks, and electrical shock. This would include wearing welder's gloves and a protective face shield and may include arm protectors, apron, hats, shoulder protection, as well as dark, non-synthetic, substantial clothing. See ANSI Z49.1. Welders should be trained not to allow electrically live parts to contact the skin or wet clothing and gloves. The welders should insulate themselves from the work and ground and should not touch live electrical parts. Welders should not wear short sleeve shirts or short pants.

· **Limitation and supervision of exposure into the environment:** None

* 9 Physical and Chemical Properties

· **Information on basic physical and chemical properties**

· **General Information**

· **Physical state**

Solid

· **Color:**

Silver/gray metallic color

· **Odor:**

Odorless until used

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· Odor threshold:	Not determined.
· Melting point/Melting range:	Not determined.
· Boiling point/Boiling range:	Not determined.
· Flammability:	Not determined.
· Explosion limits:	
· Lower:	Not determined.
· Upper:	Not determined.
· Flash point:	None
· Auto igniting:	Not applicable
· Decomposition temperature:	Not determined.
· pH-value:	Not applicable.
· Viscosity:	
· Kinematic:	Not applicable.
· Dynamic:	Not applicable.
· Solubility in / Miscibility with:	
· Water:	Soluble.
· Partition coefficient (n-octanol/water):	Not determined.
· Vapor pressure:	Not applicable.
· Vapor pressure:	
· Density:	Not determined.
· Relative density:	Not determined.
· Vapor density:	Not applicable.
· Particle characteristics	Not determined.
· Other information:	
· Appearance:	
· Form:	Solid Wire/Rod
· Important information on protection of health and environment, and on safety.	
· Ignition temperature:	Product is not self-igniting.
· Danger of explosion:	Product does not present an explosion hazard.
· Solvent content:	
· VOC content:	0.00 %
· Solids content:	100.0 %
· Change in condition	
· Evaporation rate:	Not applicable.

10 Stability and Reactivity

- **Reactivity:**
Stable under normal conditions.
May react violently or explosively on contact with water. Will react with water or steam to product hydrogen. Incompatible (violent reactions) with chlorine, fluorine, oxidizers, calcium, carbide, alkali carbonates, iodine pentafluoride, cobaltic fluoride, rubidium carbide, MnF₃, nitrosyl fluoride, AgF. Mixtures of cesium acetylide with silicon react vigorously on heating. Rubidium acetylide reacts vigorously with silicon on warming.
- **Chemical stability:** Stable under normal conditions.
- **Thermal decomposition / conditions to be avoided:** No decomposition if used according to specifications.
- **Possibility of hazardous reactions:**
May react violently or explosively on contact with water. Will react with water or steam to product hydrogen. Incompatible (violent reactions) with chlorine, fluorine, oxidizers, calcium, carbide, alkali carbonates, iodine pentafluoride, cobaltic fluoride, rubidium carbide, MnF₃, nitrosyl fluoride, AgF. Mixtures of cesium acetylide with silicon react vigorously on heating. Rubidium acetylide reacts vigorously with silicon on warming.
Contact with acids or strong bases may cause generation of gas.

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· **Conditions to avoid:** No further relevant information available.

· **Incompatible materials:**

Incompatible (violent reactions) with chlorine, fluorine, oxidizers, calcium, carbide, alkali carbonates, iodine pentafluoride, cobaltic fluoride, rubidium carbide, MnF₃, nitrosyl fluoride, AgF.

Strong acids, strong bases, strong oxidizing agents and strong reducing agents.

· **Hazardous decomposition products:**

Toxic chromium oxide fumes.

Welding fumes and gases cannot be classified simply. The composition and quantity of both are dependent upon the metal being welded, the processes and procedures followed, and the welding consumables used. Other conditions that also influence the composition and quantity of fumes and gases to which workers may be exposed include: coatings on the metal being welded (such as paint, plating, or galvanizing), the number of welders in operation and the volume of the work area, the quality and amount of ventilation, the position of the welder's head with respect to the fume plume, and the presence of contaminants in the atmosphere (such as chlorinated hydrocarbon vapors from cleaning and degreasing procedures). When the electrode is consumed, the fume and gas decomposition products generated are different in percent and form from the ingredients listed in Section 8. Fume and gas decomposition, and not the ingredients in the electrode, are important. The concentration of a given fume or gas component may decrease or increase by many times the original concentration. Also, new compounds not in the electrodes may form. The known gases and fumes that may form during welding or cutting and their exposure limits are noted in the list in Section 11 below. Decomposition products of normal operation include those originating from the volatilization, reaction, or oxidation of the materials shown in Section 8, plus those from the base metal and coating, etc. as noted above. Chlorinated solvents may be decomposed into toxic gases such as phosgene.

It is understood, however, that the elements and/or oxides to be mentioned are virtually always present as complex oxides and not as metals (See "Characterization of Arc Welding Fume", from the American Welding Society). The elements or oxides listed Section 8 correspond to the ACGIH categories found in "Threshold Limit Values for Chemical Substances and Physical Agents" listed in Section 8. Some products will also contain: carbon, chromium, copper, iron, manganese, molybdenum, nickel, niobium, silicon, tungsten and cobalt. Some elements or compounds may exceed their PELs/TLVs before the total fumes exceed 5 mg/m³.

· **Additional information:**

Niobium metal is rapidly dissolved by hydrofluoric acid or hydrofluoric-nitric acid mixtures. Niobium ignites in cold fluorine and above 200°C will react exothermically with chlorine, bromine and halocarbons such as carbon tetrachloride, carbon tetra fluoride and Freon's.

* 11 Toxicological Information

· **Information on toxicological effects:**

Effects of Over-Exposure: Electric arc welding may create one or more of the following health hazards:

- ☐ ARC RAYS can injure eyes and burn skin. Incidences of skin cancer have been reported.
- ☐ ELECTRIC SHOCK can kill.
- ☐ FUMES AND GASES GENERATED FROM WELDING can be dangerous to your health.
- ☐ PRIMARY ROUTES OF ENTRY are the respiratory system, eyes, skin, and/or indigestion.
- ☐ NOISE can damage hearing.

Short-term (acute) over-exposure effects:

- ☐ WELDING FUMES may result in discomfort, such as dizziness, nausea, or dryness or irritation of the nose, throat, or eyes.
- ☐ ALUMINUM OXIDE may cause irritation of the respiratory system.
- ☐ IRON, IRON OXIDE have no known effects. Treat as a nuisance dust or fume.
- ☐ MANGANESE, MANGANESE COMPOUNDS may cause metal fume fever, characterized by irritation of the throat, vomiting, nausea, fever, body aches, and chills. Recovery is generally complete within 48 hours of overexposure.
- ☐ MOLYBDENUM may cause irritation of the eyes, nose, and throat.
- ☐ NICKEL, NICKEL COMPOUNDS may cause metallic taste, nausea, tightness in chest, fever, and allergic reactions.

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- ☐ TITANIUM DIOXIDE may cause irritation of the respiratory system.
- ☐ COPPER may cause capillary damage, headache, cold sweat, weak pulse, and kidney and liver damage, central nervous system excitation followed by depression, jaundice, convulsions, paralysis, and coma. Death may occur from shock or renal failure.

Long-term (chronic) over-exposure effects:

- ☐ WELDING FUMES in excess levels may cause bronchial asthma, lung fibrosis, pneumoconiosis, or 'siderosis.' Overexposure to air contaminants may lead to their accumulation in the lungs, a condition which may be seen as dense areas on chest x-rays. The severity of the change is proportional to the length of exposure. The changes seen are not necessarily associated with symptoms or signs of reduced lung function or disease. In addition, the changes on X-rays may be caused by non-work factors such as smoking, etc.
- ☐ ALUMINUM OXIDE may cause pulmonary fibrosis and emphysema.
- ☐ IRON, IRON OXIDE may cause siderosis or deposits of iron in the lungs, which is believed to affect pulmonary function. Lungs will clear in time when exposure to iron fumes and its compounds ceases. Iron and magnetite (Fe₃O₄) are not regarded as fibrogenic materials.
- ☐ MANGANESE, MANGANESE COMPOUNDS may cause central nervous system effects referred to as 'manganism.' Symptoms include languor, sleepiness, muscular weakness, emotional disturbances, spastic gait, and tremors. Behavioral changes and changes in handwriting may also appear. These effects are irreversible. Employees overexposed to manganese should receive regular medical examinations for early detection of manganism.
- ☐ MOLYBDENUM prolonged overexposure may result in loss of appetite, weight loss, loss of muscle coordination, difficulty in breathing, and anemia.
- ☐ NICKEL, NICKEL COMPOUNDS may lung fibrosis or pneumoconiosis. Studies of nickel refinery workers indicated a higher incidence of lung and nasal cancers.
- ☐ TITANIUM DIOXIDE may cause pulmonary irritation and slight fibrosis.
- ☐ COPPER may cause hepatic cirrhosis, brain damage and demyelination, kidney defects, and copper deposition in the cornea as exemplified by humans with Wilson's disease. It has also been reported that copper poisoning has led to hemolytic anemia and accelerates arteriosclerosis.

Acute toxicity:

LD/LC50 values that are relevant for classification:

7439-89-6 Iron		
Oral	LD50	7,500 mg/kg (Rat)
7440-47-3 Chromium		
Inhalative	LC50/96 hours	14.3 mg/l (Cyprinus carpio)
7439-96-5 Manganese		
Oral	LD50	9,000 mg/kg (Rat)
7439-98-7 Molybdenum		
Oral	LD50	>5,000 mg/kg (Rat)
Dermal	LD50	>2,000 mg/kg (Rat)
Inhalative	LC50/4 h	800 mg/l (Trout)
		>5.84 mg/l (Rat)
7440-21-3 Silicon		
Oral	LD50	3,160 mg/kg (Rat)
7440-33-7 Tungsten		
Oral	LD50	2,000 mg/kg (Rat)
Dermal	LD50	2,000 mg/kg (Rat)
Inhalative	LC50/4 h	5.4 mg/l (Rat)

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7440-03-1 Niobium		
Oral	Toxic Dose Low	>10,000,000 µg/kg (mouse) >10,000,000 µg/kg (Rat)
7440-48-4 Cobalt		
Oral	LD50	6,170 mg/kg (Rat)

· **Primary irritant effect:**

· **On the skin:**

Irritant to skin and mucous membranes.

May cause an allergic skin reaction.

· **On the eye:** Irritating effect.

· **Sensitization:**

Sensitization possible through inhalation.

Sensitization possible through skin contact.

· **Additional toxicological information:**

The product shows the following dangers according to internally approved calculation methods for preparations:

Harmful

Irritant

Symptoms of systemic copper poisoning may include: capillary damage, headache, cold sweat, weak pulse, and kidney and liver damage, central nervous system excitation followed by depression, jaundice, convulsions, paralysis, and coma. Death may occur from shock or renal failure. Chronic copper poisoning is typified by hepatic cirrhosis, brain damage and demyelination, kidney defects, and copper deposition in the cornea as exemplified by humans with Wilson's disease. It has also been reported that copper poisoning has lead to hemolytic anemia and accelerates arteriosclerosis.

· **Interactive effects** No interactive effects between components are known.

· **Carcinogenic categories:**

· **IARC (International Agency for Research on Cancer):**

Group 1 - Carcinogenic to humans

Group 2A - Probably carcinogenic to humans

Group 2B - Possibly carcinogenic to humans

Group 3 - Not classifiable as to its carcinogenicity to humans

Group 4 - Probably not carcinogenic to humans

7440-47-3	Chromium	3
7440-02-0	Nickel	2B
7440-48-4	Cobalt	2A

· **NTP (National Toxicology Program):**

7440-02-0	Nickel	R
7440-48-4	Cobalt	R

· **OSHA-Ca (Occupational Safety & Health Administration):**

None of the ingredients are listed.

· **Alternative sources for toxicological information**

No non-standard sources for toxicological information where used.

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12 Ecological Information

· **Toxicity:**

· **Aquatic toxicity:**

7440-47-3 Chromium

EC50 0.07 mg/l (Water flea)

7440-02-0 Nickel

EC50 1 mg/l (Water flea)

7439-96-5 Manganese

EC50 40 mg/l (Water flea)

7440-50-8 copper

EC50 0.04-0.05 mg/l (Water flea)

· **Persistence and degradability:** No further relevant information available.

· **Bioaccumulative potential:** No further relevant information available.

· **Mobility in soil:** No further relevant information available.

· **Results of PBT and vPvB assessment:**

· **PBT:** Not applicable.

· **vPvB:** Not applicable.

· **Other adverse effects:**

· **Additional ecological information:**

· **General notes:**

Do not allow product to reach ground water, water course or sewage system.

Danger to drinking water if even small quantities leak into the ground.

* 13 Disposal Considerations

· **Waste treatment methods**

· **Recommendation:**

Must not be disposed of together with household waste. Do not allow product to reach sewage system.

Observe all federal, state and local environmental regulations when disposing of this material.

· **Uncleaned packaging**

· **Recommendation:** Disposal must be made according to official regulations.

· **Recommended cleansing agent:** Water, if necessary with cleansing agents.

14 Transport Information

· **UN-Number:**

· **DOT, ADR/ADN, ADN, IMDG, IATA**

Non-Regulated Material

· **UN proper shipping name:**

· **DOT, ADR/ADN, ADN, IMDG, IATA**

Non-Regulated Material

· **Transport hazard class(es):**

· **DOT, ADR/ADN, ADN, IMDG, IATA**

· **Class:**

Non-Regulated Material

· **Packing group:**

· **DOT, ADR/ADN, IMDG, IATA**

Non-Regulated Material

· **Environmental hazards:**

Not applicable.

· **Transport in bulk according to Annex II of**

MARPOL73/78 and the IBC Code:

Not applicable.

· **Special precautions for user:**

Not applicable.

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· **UN "Model Regulation":** Non-Regulated Material

15 Regulatory Information

- **Safety, health and environmental regulations/legislation specific for the substance or mixture:**
- **SARA (Superfund Amendments and Reauthorization):**

· **Section 355 (extremely hazardous substances):**

None of the ingredients are listed.

· **Section 313 (Specific toxic chemical listings):**

7440-47-3 Chromium

7440-02-0 Nickel

7439-96-5 Manganese

7440-50-8 copper

7440-48-4 Cobalt

· **TSCA (Toxic Substances Control Act):**

All components have the value ACTIVE.

· **Hazardous Air Pollutants**

7439-96-5 Manganese

7440-48-4 Cobalt

· **California Proposition 65:**



WARNING: This product can expose you to chemicals including the listed chemicals which are known to the State of California to cause cancer, birth defects and/or other reproductive harm. For more information, go to www.P65Warnings.ca.gov.

· **Chemicals known to cause cancer:**

7440-02-0 Nickel

7440-48-4 Cobalt

· **Chemicals known to cause reproductive toxicity for females:**

None of the ingredients are listed.

· **Chemicals known to cause reproductive toxicity for males:**

None of the ingredients are listed.

· **Chemicals known to cause developmental toxicity:**

None of the ingredients are listed.

· **New Jersey Right-to-Know List:**

7440-47-3 Chromium

7440-02-0 Nickel

7439-96-5 Manganese

7439-98-7 Molybdenum

7440-50-8 copper

7440-21-3 Silicon

7440-33-7 Tungsten

7440-48-4 Cobalt

· **New Jersey Special Hazardous Substance List:**

7440-47-3 Chromium

F3

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7440-02-0	Nickel	CA
7439-96-5	Manganese	F3, R1
7440-21-3	Silicon	F3
7440-33-7	Tungsten	F3
7440-48-4	Cobalt	CA, F3

· Pennsylvania Right-to-Know List:

7440-47-3	Chromium	
7440-02-0	Nickel	
7439-96-5	Manganese	
7439-98-7	Molybdenum	
7440-50-8	copper	
7440-21-3	Silicon	
7440-33-7	Tungsten	
7440-48-4	Cobalt	

· Pennsylvania Special Hazardous Substance List:

7440-47-3	Chromium	ES
7440-02-0	Nickel	ES
7439-96-5	Manganese	E
7440-50-8	copper	E
7440-48-4	Cobalt	E

· Carcinogenic categories:
· EPA (Environmental Protection Agency):

7440-47-3	Chromium	D
7439-96-5	Manganese	D
7440-50-8	copper	D

· TLV (Threshold Limit Value established by ACGIH):

7440-47-3	Chromium	A4
7440-02-0	Nickel	A5
7439-98-7	Molybdenum	A3
7440-48-4	Cobalt	A3

· NIOSH-Ca (National Institute for Occupational Safety and Health):

7440-02-0	Nickel	
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· GHS label elements

The product is classified and labeled according to the Globally Harmonized System (GHS).

· Hazard pictograms:

· Signal word: Danger
· Hazard-determining components of labeling:

Nickel
Iron

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Cobalt
copper

· **Hazard statements:**

H315 Causes skin irritation.
H319 Causes serious eye irritation.
H334 May cause allergy or asthma symptoms or breathing difficulties if inhaled.
H317 May cause an allergic skin reaction.
H351 Suspected of causing cancer. Route of exposure: Inhalation.
H335 May cause respiratory irritation.
H372 Causes damage to the respiratory system through prolonged or repeated exposure. Route of exposure: Inhalation.

· **Precautionary statements:**

P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P260 Do not breathe dust/fume/gas/mist/vapors/spray.
P264 Wash thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P272 Contaminated work clothing must not be allowed out of the workplace.
P280 Wear protective gloves/protective clothing/eye protection/face protection/hearing protection.

P284 [In case of inadequate ventilation] wear respiratory protection.
P302+P352 If on skin: Wash with plenty of water.
P304+P340 If inhaled: Remove person to fresh air and keep comfortable for breathing.
P305+P351+P338 If in eyes: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P308+P313 IF exposed or concerned: Get medical advice/attention.
P312 Call a poison center/doctor if you feel unwell.
P321 Specific treatment (see supplementary first aid instructions on this Safety Data Sheet).
P362+P364 Take off contaminated clothing and wash it before reuse.
P333+P313 If skin irritation or rash occurs: Get medical advice/attention.
P337+P313 If eye irritation persists: Get medical advice/attention.
P342+P311 If experiencing respiratory symptoms: Call a poison center/doctor.
P403+P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.
P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

· **National regulations:**

The product is not subject to be labelled according with the prevailing version of the regulations on hazardous substances.

· **Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

16 Other Information

SOWESCO urges each end user and recipient of this SDS to study it carefully. If necessary, consult an industrial hygienist or other expert to understand this information and safeguard the environment and protect workers from potential hazards associated with the handling or use of this product. This information is believed to be accurate as of the revision date shown above. However, no warranty, expressed or implied, is given. Because the conditions or methods of use are beyond SOWESCO's control, we assume no liability resulting from the use of this product. Regulatory requirements are subject to change and may differ between various locations. Compliance with all applicable Federal, State, Provincial, and Local laws and regulations remain the responsibility of the user.

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- **Date of previous version** 09/14/2022
- **Version number of previous version:** 3
- **Date of preparation** 06/23/2025
- **Abbreviations and acronyms:**

ADR: The European Agreement concerning the International Carriage of Dangerous Goods by Road
 ADN: The European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways
 IMDG: International Maritime Code for Dangerous Goods
 DOT: US Department of Transportation
 IATA: International Air Transport Association
 EINECS: European Inventory of Existing Commercial Chemical Substances
 ELINCS: European List of Notified Chemical Substances
 CAS: Chemical Abstracts Service (division of the American Chemical Society)
 NFPA: National Fire Protection Association (USA)
 HMIS: Hazardous Materials Identification System (USA)
 VOC: Volatile Organic Compounds (USA, EU)
 LC50: Lethal concentration, 50 percent
 LD50: Lethal dose, 50 percent
 PBT: Persistent, Bioaccumulative and Toxic
 vPvB: very Persistent and very Bioaccumulative
 NIOSH: National Institute for Occupational Safety and Health
 OSHA: Occupational Safety & Health Administration
 TLV: Threshold Limit Value
 PEL: Permissible Exposure Limit
 REL: Recommended Exposure Limit
 BEI: Biological Exposure Limit
 Flammable solids 1: Flammable solids – Category 1
 Flammable solids 2: Flammable solids – Category 2
 Pyrophoric solids 1: Pyrophoric solids – Category 1
 Substances and mixtures which, in contact with water, emit flammable gases 1: Substances and mixtures which in contact with water emit flammable gases – Category 1
 Acute toxicity - oral 4: Acute toxicity – Category 4
 Skin irritation 2: Skin corrosion/irritation – Category 2
 Eye irritation 2A: Serious eye damage/eye irritation – Category 2A
 Eye irritation 2B: Serious eye damage/eye irritation – Category 2B
 Sensitization - respiratory 1: Respiratory sensitisation – Category 1
 Sensitization - skin 1: Skin sensitisation – Category 1
 Germ cell mutagenicity 2: Germ cell mutagenicity – Category 2
 Carcinogenicity 1B: Carcinogenicity – Category 1B
 Carcinogenicity 2: Carcinogenicity – Category 2
 Reproductive toxicity 1B: Reproductive toxicity – Category 1B
 Specific target organ toxicity (single exposure) 3: Specific target organ toxicity (single exposure) – Category 3
 Specific target organ toxicity (repeated exposure) 1: Specific target organ toxicity (repeated exposure) – Category 1
 Aquatic Acute 1: Hazardous to the aquatic environment - acute aquatic hazard – Category 1
 Aquatic Acute 3: Hazardous to the aquatic environment - acute aquatic hazard – Category 3

- *** Data compared to the previous version altered.**

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